MA1521 L1 Week 10 Wed 20261021

Convert $\int_0^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{-(x^2+y^2)}^{(x^2+y^2)} 21xy^2 \, dz \, dy \, dx$ to cylindrical coordinates and evaluate the new integral.

Convert $\int_0^{2\pi}\int_0^{\sqrt{2}}\int_r^{\sqrt{4-r^2}} 3\,dz\,r\,dr\,d\theta$, to rectangular coordinates with the order of integration $dz\,dx\,dy$ and evaluate the new integral.

Evaluate: $\iiint_B z\,dx\,dy\,dz$ where B is the region within the cylinder $x^2 + y^2 = 1$
above the xy plane
and below the cone $z = (x^2 + y^2)^{1/2}$.

Evaluate: $\iiint_W (x^2 + y^2 + z^2)^{-1/2}\, dx\, dy\, dz$ where W is the region determined by $\frac{1}{2} \leq z \leq 1$ and $x^2 + y^2 + z^2 \leq 1$.